

Chp #1

Theory of demand

Topic #2:- Utility, price and demand.



Zarek Mushtaq

Concept of Equi-marginal principle

2 goods $\rightarrow$ consumption

$\downarrow$
e.g. good Y

Price_Y marginal utility_Y

i.e. $\frac{\text{marginal utility of good Y}}{\text{Price of good Y}}$

As there are 2 goods, therefore

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$$

Fall in price of good X, $Q_{D_X} \uparrow$

$$\frac{MU_X}{P_X} > \frac{MU_Y}{P_Y}$$

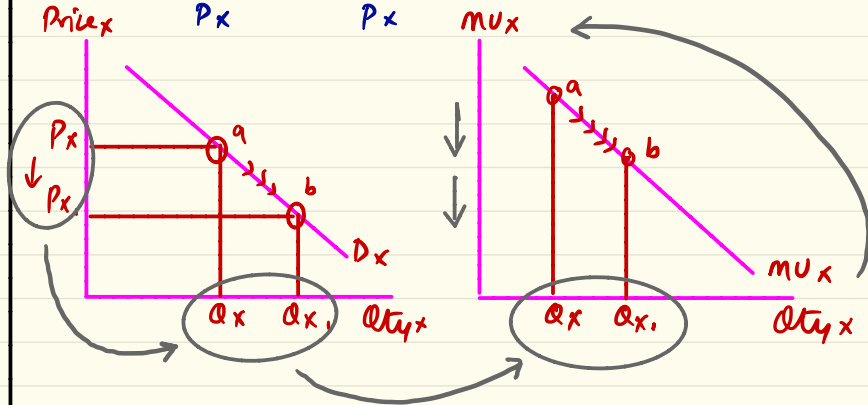
Diminishing marginal utility of good X
Switch to good Y

$$\frac{MU_Y}{P_Y} > \frac{MU_X}{P_X}$$

Diminishing Marginal utility of good Y

A rational consumer will keep on substituting unless

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$$



Price changes $\Rightarrow$ affects quantity

Thus price affects marginal utility!

$\downarrow$
changes in quantity
 $\downarrow$
affects marginal utility

Connection of utility, price and demand :-

Thus we have attained the normal demand relationship that other things being equal, as the price of good x falls, more of it is bought. We have therefore, a normal downward sloping demand curve i.e. the theory of diminishing marginal utility and the theory of equal MU_s per $\$$ spent gives a negative relationship between quantity demanded of a commodity and its price, so the individual demand curve for any good will be same as their marginal utility curve where utility is measured in monetary terms.

One commodity equilibrium :- If we hold the consumption of other products constant, we can study consumer behaviour and derive the demand curve from marginal utility schedule. Let the consumer be faced with the given market price of some product. The price shows what he must sacrifice to obtain each unit of the product. The marginal utility schedule or graph shows the value he places on each unit of the product. Clearly the consumer can increase his utility by adding to his purchases as long as the value he places on each additional unit exceeds the amount he must pay for that unit. It follows that the utility maximizing consumer will adjust his purchases of product until the marginal utility of the last unit purchased is equal to the price of that product i.e. $MU_x = P_x$

MULTI-COMMODITY EQUILIBRIUM: Assume that the consumer has a choice between two products X and Y both cost \$1 each and consumer has \$1 to spend then obviously the commodity which yields the greater utility will be bought.

If we were to apply this principle to each successive unit of the consumer's spending, we could conclude that utility will be maximized when income has been allocated in such a way that the utility derived from one extra dollar's worth of X is equal to the utility derived from the consumption of one extra dollar's worth of Y. If this calculation is not fulfilled then the consumer could obviously increase the total utility by switching the expenditures from X to Y or vice versa. For example, a consumer would not spend a dollar extra more on vegetables if it was considered that the same dollar spent on meat would give more satisfaction or in the reverse situation would not spend a dollar extra on meat if he considers that the same dollar spent on vegetables would yield greater satisfaction.

The extra satisfaction derived from the consumption of one more unit of X is its MU which we can write as MU_x and that of Y as MU_y . We must also consider the relative prices of one more unit of X and Y which we can write as P_x and P_y . We can then see that the utility maximizing condition when:

$$MU_x/P_x = MU_y/P_y$$

Suppose that, starting from a condition of equilibrium, the price of X falls relative to Y. We now have a condition where the utility from the last dollar spent on X will be greater than the utility from the last dollar spent on Y.

$$MU_x/P_x > MU_y/P_y$$

Multi-commodity equilibrium /
Equi-marginal principle /
Consumer equilibrium

$$\left[\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \right]$$

- 24 Rational consumer Romesh spent 90 rupees buying herb X and spice Y. Both cost 10 rupees per gram.

grams purchased	total utility herb X	total utility spice Y
2	15	12
3	30	25
4	45	35
5	55	42

Ans:- D

How much of each should he buy?

	herb X (grams)	spice Y (grams)
A	2	3
B	3	5
C	4	5
D	5	4

- 37 The price of good S is \$6 and the price of good T is \$8. A consumer's marginal utility for each good is shown.

units	marginal utility of good S	marginal utility of good T
1	36	32
2	24	22
3	18	20
4	9	18
5	0	9

Ans:- B

If this consumer wants to maximise total utility, which combination of goods would they consume?

	good S	good T
A	1	1
B	2	1
C	3	4
D	5	5

Questions

→ Use the screen shot!

Q1 Explain equi-marginal principle? 6 (Use ratios)

Q2 Explain relationship between utility, price and demand? 6
(Paragraph given)
↳ Thus we have....